

LLM 2-oy 2-Dave Report.

Git - loyihada organizatsiya tarkibini saqlash beradigan dastur.

GitHub - loyihani Internetda saqlash uchun foylaniladigan platforma.

Gitignore - loyihadagi ko'rsatilgan fayl va papkalarni GitHubga yubormaslik.

Bizda loyihani production qilgach uni monitoring va managing qilish zarur bo'ladi. Boshqarish va kuzatish bo'yicha sarfay amallarga kiradi.

gitignore
Bizda hatta fayllarni va kerak bo'lmagan fayl papkalarni yubormaslik kerak.

- Massive model weights
- Dataset
- Token / API (private)

Gitignore ichidagi qoidalar

- Ignore massive data files
- Ignore large model weights va checkpoints
- Ignore python environment and Local Virtual Env
- Ignore Jupyter Notebook Configuration cache and System files
- Ignore Hugging Face and tokenisation caches

SM L dagi gitignore b-n LLM dagi o'xshash 3 ta farg mavjud

- ① False soc impact - LLM da false
dilehami juda katta
- ② Dataset tracking - LLM da text dan
borat
- ③ Cost of Leaks - molyapdy chiqoulin
katta

Data Sources.

Qanaqa LLM yaratmoqchimisiz?

- ① General purpose
- ② Special shunga qarab data topamiz

General Purpose uchun Data Source

- ① commoncrawl.org
- ② huggingface.co/datasets/HuggingFaceFW/fineweb
- ③ github.com/allenai/dolma

Mashxur LLM qilinadigan sourcelar

Common Crawl - web

FineWeb - web text

FineWeb-Edu - education

The Pile - multi domain

Red Pajama - multi domain

Dolma - multi domain

The Stack v2 - code

arXiv - Academic

Qanday tanlash haqqida qarash

Best choices for Our Student Project

General SLM

- Fine web

Education SLM

- Fine web - Edu

Code SLM

- The Stack v2

Scientific SLM

- OMD

↓ Build our own SLM ↓

① Choose Dataset

Fine web

Fine web Edu

The Stack

② Download small subset

③ clean data

④ deduplicate

⑤ tokenization

⑥ create token dataset

⑦ Train SLM

⑧ Evaluate

⑨ Instruction Fine tuning

⑩ Deploy SLM

Quality Dimension - datasetni yaxshi

yoki zamonaviy uni sifatidan bilar

Qaydagi ta'lim bo'yicha dataset
baholab kelishi kerak

Relevance

bu qaydagi loyihalar
bo'yidagi mas'uliyatlar

Quality -

dataning sifati yaxshimi
yoki bekorchi malemi?

Diversity - bir qat'iyotda bir qat'iyotda
 uchun barcha tarmoqlar
 qamrab olgani uchun tarmoqlar
 emas barcha tarmoqlar
 bo'lgani afzal.

Data Mixture

SPM da dataset foydalanishda
 bo'lgani uchun shu bilan fransuz

Shunda LLM da dataset analizi
 bo'lgani yaxshi

Source	Example Domain	Example Size
Free Web	Web	50X
Wikipedia	Reference	10X
Free Web Edu	Education	15X
arXiv	Science	10X
Code dataset	Programming	10X
Books / public domain text	Books	5X

Alma uchun mixture - chunki surti
 xal dataset bu surti xal ilmi degan

Data mixture qolganda Random
 Combination qilinmaydi.

Web = 30X
 Code = 5X
 Science = 5X

bu xato

Idea: The data mixture is one of the
 model design decision.

Licence

- who created it?
- where did the data come from?
- what licence applies?
- what are the conditions?
- can I modify it?
- can I redistribute it?
- can I use it commercially?
- Are there additional upstream terms?

Licence types: GPL - CC0, CC BY, CC BY-NC, CC BY-NC-SA, CC BY-ND, CC BY-NC-ND, ODC-BY, ODbL, MIT, Apache-2.0, BSD-3, CPL-3.0

Licence (game) faylga yoxib qayt
 data sets: variance / not licence data
 evidence: with
 source - code generation

Design LLM

1. Soha tanlanadi (Generalist, educational, the tanlanadi)
2. LLM nomi yoziladi
3. Main purpose - naif
4. Target user boydalanadi
5. Source yoziladi
6. Dataset yoziladi
7. Quality yoziladi
8. Evaluate yoziladi
9. Data yoziladi
10. Check yoziladi
11. Design yoziladi
12. Final decision